



SEMESTER 1
2022-2023

CS370FZ
Computation and Complexity

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Time allowed: 2 hours

Answer all **three** questions

All three questions carry equal marks

Instructions

	Yes	No
Log Books Allowed	☐	☒
Formula Tables Allowed	☐	☒
Other Allowed (<i>enter details</i>)	☐	☒

General (*enter details*)

- 1** **[25 marks]**
- (a) Give self-contained definitions of the following languages/sets: [12 marks]
- A_{TM} (the acceptance problem),
 - $HALT_{TM}$ (the halting problem),
 - TM M decides Language L ,
 - TM M semidecides Language L .
- (b) Show that $HALT_{TM}$ is undecidable. (Assume we know A_{TM} is undecidable, you are required to reduce A_{TM} to $HALT_{TM}$.) [5 marks]
- (c) Show that a language is decidable iff it is Turing-recognizable and co-Turing-recognizable. [8 marks]
- 2** **[25 marks]**
- (a) State Rice's Theorem. [5 marks]
- (b) Prove Rice's Theorem (you can use the recursion theorem). [8 marks]
- (c) Give self-contained definitions of the following concepts: [6 marks]
- an oracle TM,
 - Language A is Turing reducible to language B .
- (d) Show that E_{TM} is Turing reducible to A_{TM} (E_{TM} is the emptiness testing problem for TMs). [6 marks]
- 3** **[25 marks]**
- (a) Give the definitions of the following classes: [4 marks]
- Class P ,
 - Class NP .
- (b) Give the definitions of the following sets. Your definition should be self-contained (e.g. for $CLIQUE$, explain what a k -clique is). [6 marks]
- $HAMPATH$,
 - SAT ,
 - $COMPOSITES$.
- (c) Show that $PATH \in P$. [5 marks]
- (d) Recall the proof $3SAT$ is polynomial time mapping reducible to $CLIQUE$ ($3SAT \leq_P CLIQUE$). From the following ϕ construct the associated graph G . Explain how to obtain a 3-clique for G from a satisfying assignment for ϕ . [10 marks]
- $$\phi = (x_1 \vee \bar{x}_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2 \vee x_2)$$